

Technical Research Report

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	03-09-2026	Draft Version	Casey Stijlaart	Draft Version	Initial document layout, non-camera sensing techniques chapter
0.2	04-09-2026	Draft Version	Casey Stijlaart	Draft Version	draft version of sensing techniques
0.3	07-09-2026	Draft Version	Casey Stijlaart	Draft Version	Finalisation of non-camera sensing techniques

2. Non-Camera Person Position & Height Sensing Techniques

2.1. Introduction

This chapter addresses the first technical research question defined in the Project Plan (Chapter 3.6, Research Questions), referred to as **RQ1**: "Which combination of non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?" Two further sub-questions are referenced by number: **RQ2** (robust, low-latency position estimates under real-world noise) and **RQ4** (extensibility to multi-person scenarios); **RQ3** (on-device ML for height classification) is out of scope here.

Per the Project Plan (Chapter 3.3, Assignment), the target system estimates a person's real-time 3D position and height within the Holobox space without cameras, for the Virtual Human to act on. This chapter surveys candidate technologies against the criteria they must satisfy, compares current hardware and cost, and recommends a sensor choice with a staged de-risking plan rather than a final hardware order, since a fully polished prototype is out of scope (Chapter 3.4, Scope and Preconditions). A general height estimate is treated as sufficient for now, and multi-person tracking is addressed only as an evaluation criterion, not a capability this stage builds.

2.2. Research Approach

This chapter follows the Library and Lab strategies of the DOT-framework identified for this research question in the Project Plan (Chapter 3.7, Research Methods): available product analysis, literature study, document analysis, and bench experimentation.

2.2.1. Library: Literature Study

Candidate Sensor Technologies gathers and verifies peer-reviewed literature for each non-camera sensing family against RQ1's position/height requirement and, separately, against RQ4's multi-person active-speaker problem, in two separate passes: RQ1 sits in the radar/thermal/RF position-and-pose-estimation literature, while RQ4 sits closer to acoustic and vibrometry-based speaker localisation, so a single combined search would have missed sources specific to either.

2.2.2. Lab: Available Product Analysis

Alongside the literature, the same chapter narrows the literature-backed technology families down to specific, currently sourceable hardware modules, checked against live distributor listings (Digi-Key, Mouser, Adafruit, and others) rather than assumed pricing, since an unverified cost estimate is not a usable basis for a hardware decision or the budget-approval process described in the Project Plan (Chapter 6.1, Finance and Risks). Literature backing and product availability are two different questions: a technique can be well-published without there being a specific, currently purchasable module that implements it at a reasonable price, or at all.

2.2.3. Lab: Bench Experimentation Strategy

The recommended path into bench experimentation commits to mmWave radar directly, managing implementation risk by splitting the build into capability-based phases rather than assuming the full pipeline works at once: position tracking first, then height estimation, then multi-person clustering, then speaker detection, each building on an already-working previous phase. This is deliberate: it isolates a failure to whichever capability is currently being added, rather than letting it surface somewhere in an untested, fully assembled system; the same mmWave hardware and UART/point-cloud/Unreal Engine bridge built in phase one is reused, not redone, in every phase after it; it follows the research-question sequence, validating RQ1 (position/height) before adding RQ4 (multi-person clustering, speaker detection); and it reflects feature priority, since RQ4's extensions improve the system without being required for the current single-person scope. A working fallback exists at each stage: if a later phase's capability underperforms, the earlier phase's version remains a functioning system rather than a dead end, mirroring the fallback the project's own risk register anticipates for both sensor selection and on-device ML performance.

2.3. Evaluation Criteria

Before diving into available sensor modules, it is of great importance to understand the requirements they need to meet. Thus the comparison can be set out against explicit, justified requirements rather than an implicit sense of what "good" looks like. Each criterion below is traced to either a technical sub-question described in the Project Plan (Chapter 3.6, Research Questions), a key feature named in the Project Plan (Chapter 3.3, Assignment), or a project constraint from the Project Plan (Chapter 6.1, Finance and Risks), rather than to a generic notion of sensor quality.

Table 2. Evaluation Criteria

Criterion	Description
Height-estimation capability	Must produce an actual height reading (alone or fused), not just (x, y) position, per RQ1's spatial-resolution requirement; a general estimate suffices for now.
Position accuracy and coverage	Position error and range must be usable at the Holobox's actual room-scale dimensions, since RQ1 covers (x, y) as well as height and a sensor that can't cover the space isn't usable regardless of height capability.
Real-time embedded feasibility	Interactive latency on an embedded platform (targeting <100 ms, hard ceiling ~1 s), as RQ2 requires low-latency embedded estimates; thresholds follow Nielsen's (1993) response-time limits, building on Miller (1968).
Multi-person extensibility	Shouldn't fundamentally break if a second person enters the space later, since RQ4 and the Assignment both name multi-person as a future-facing requirement.
Data minimisation / privacy	Output can't identify an individual and captures only what's needed for position and height, required by the ethical/legal sub-questions given that visitors are children aged 8-12 (AVG/GDPR).
Literature and community backing	Prior published work must show the specific capability required, not just the technology in general, reducing the risk of committing project time to proving something never demonstrated at all.
Hardware cost	Cost of the sensor and any paired second sensor, checked against sourceable listings; a real constraint since no project budget has yet been agreed with MindLabs.

These criteria are not weighted equally, and are not applied as a hard pass/fail filter. Height-estimation capability and data minimisation are treated as closer to hard requirements, since they follow directly from RQ1 and the ethical/legal sub-questions; the remaining criteria (sensor combination, cost, power/footprint) are weighed against each other in the chapter that follows, since a technology that is merely more expensive or harder to pair with a second sensor is not disqualified the way one with no height precedent or an identifying output effectively is.

2.4. Candidate Sensor Technologies

This chapter surveys non-camera sensing technologies capable of device-free person detection, localisation, and (where the literature supports it) height estimation, evaluated against the technical sub-questions described in the Project Plan (Chapter 3.6, Research Questions): primarily sufficient 3D spatial resolution for height, embedded-platform feasibility, and the data-minimisation/privacy constraints that follow from the primary visitors being children aged 8-12. A general height estimate is sufficient for now; distinguishing adult and child height categories is a possible later refinement, not a hard requirement.

Each technology is covered once, in two parts: what the literature has actually shown it can do, and, for the technically viable candidates, how that holds up against the criteria set out earlier, specific hardware, cost, and pros/cons, closing in a comparison summary, criteria matrix, and recommendation. Ngamakeur et al. (2020) frame this same comparison at a broader scale, structuring device-free sensing into a pipeline (Figure 1) used below to organise what each technology has actually been shown to do, rather than what it could in principle do.



Figure 1. Device-free Pipeline

2.4.1. Sensing Technologies

Where a technology has literature for both problems, its write-up splits into Position & Height Localisation and Speaker Identification subsections; several technologies are covered for localisation only, and Laser Doppler Vibrometry and the Microphone Array for speaker identification only. None of the speaker-identification sources have been tested in this project's specific context (children's voices, a room-scale interactive installation), so they remain a lead for further work rather than a settled result.

2.4.1.1. Time-of-Flight & LiDAR

2D LiDAR measures range by timing a laser pulse's reflection while scanning a single horizontal plane, producing an (x, y) point cloud at the unit's mounted height; well suited to in-plane tracking, but a single fixed-height plane does not by itself yield a height measurement (would need multiple units at different heights, a tilted scan, or a full 3D ToF/LiDAR unit).

Within its native (x, y) role, 2D LiDAR performs well: Bouazizi et al. (2024) fuse multiple units with a ConvLSTM model and show tolerance to furniture occlusion. For RQ4, Alam et al. (2021) document "crossover ambiguity" as 2D LiDAR's recurring multi-person failure mode when tracks pass close together, the concrete risk to plan around later. The representative module, the RPLIDAR A2M8, scans at 5-15 Hz (typically 10 Hz) per Slamtec's datasheet, the limit on full (x, y) frame availability regardless of its faster ~4000 samples/sec per-sample rate.

Pros

- Strong in-plane position accuracy and furniture-occlusion tolerance when multiple units are fused (Bouazizi et al.).
- Mature, widely available consumer/robotics hardware (RPLIDAR family) with broad community support.

Cons

- No height reading from a single scan plane; needs a second sensor or added mechanical complexity to get one.
- "Crossover ambiguity" is a documented multi-person failure mode when tracks pass close together (Alam et al.).
- 5-15 Hz scan rate limits how quickly a full (x, y) frame becomes available.

2.4.1.2. mmWave Radar

mmWave radar has an extensive body of research literature covering both position tracking and skeletal/joint-level pose estimation, and its point-cloud output is inherently non-identifying (no recognisable imagery), aligning with the sub-questions on minimising personally identifiable data from child visitors. Available product analysis narrows this to a 60 GHz point-cloud radar module built around the TI IWR6843 chip, over 24 GHz presence sensors (binary presence flag only, no point cloud) and undocumented generic modules lacking reference software. Several IWR6843 evaluation-board variants exist at different price points and field-of-view characteristics; **Module Selection** compares them and picks one to carry forward.

Position & Height Localisation

For position tracking, Huang et al. (2021) combine clutter removal, clustering, and Kalman filtering for real-time multi-person tracking on a Raspberry Pi 4, and Pegoraro and Rossi (2022) report position error down to roughly 7.6 cm. More directly relevant to height estimation is the pose-estimation literature: Zhu et al.'s ProbRadarM3F (2024) recovers 14 body keypoints from radar point clouds; since keypoints include head/shoulder joints, this is the most direct existing path to an actual height estimate rather than a size proxy, backed by a camera-free radar dataset with joint ground truth (Su et al., 2024). On latency, reported tracking papers give broadly similar tens-of-milliseconds frame periods on different radar chips (15-20 fps), and the IWR6843 chip itself supports frame rates up to ~76 Hz per TI's datasheet, though typical demo configurations run at 10-20 Hz; see Evaluation Criteria for the thresholds this is checked against.

Speaker Identification

Whether the IWR6843 reports two people as distinct targets, rather than merging them into one detection, depends on the radar's angular and range resolution, not on a single blanket "multi-person" capability. Two people need roughly 1.7-2.9m of horizontal separation, or a height difference of ~1.7m (e.g. one seated, one standing), before angular resolution reliably reports them as separate; range separation (front-back) is far finer, ~4-5cm, and follows automatically from time-of-flight rather than needing angular discrimination. Speaker identification does not share this constraint: it relies on Doppler micro-motion detection of mouth/chest movement

during speech, so the radar can identify who is speaking regardless of how close together people are standing. This vibrometry capability extends to sound itself: Xu et al.'s WaveEar (2019) uses a low-cost mmWave probe to first localise a speaker among multiple people, then steer toward their throat, solving "who is speaking" and "what are they saying" from the same sensor. For a room roughly 5m wide with 2-3 people scattered across it, ordinary spacing satisfies counting, height, and speaker-ID requirements together; the failure case to plan around is 4+ people clustered tightly, where horizontal separation drops below the 1.7-2.9m threshold and tracks become ambiguous.

Pros

- Extensive literature base: real-time multi-person tracking and keypoint-level pose/height estimation.
- Demonstrated real-time operation on a Raspberry Pi 4.
- Point-cloud output is inherently non-identifying, fitting the project's data-minimisation requirements.
- The same sensor also supports speaker identification via Doppler vibrometry, without additional hardware.

Cons

- Sensor cost is €160-185 for the IWR6843AOPEVM variant (see [Module Selection](#)).
- Height estimation still requires extra work: a blob-size heuristic or a keypoint-extraction model, not a single established method.
- No project budget has yet been agreed with MindLabs, so this cost is a real, unresolved risk.

2.4.1.3. Wi-Fi CSI Sensing

Wi-Fi Channel State Information (CSI) sensing infers presence and movement from fluctuations in the wireless channel caused by a body moving through it, using only commodity Wi-Fi hardware, but the gathered literature is consistent that it trades away spatial resolution and stability: CSI is sensitive to environmental changes (furniture, multipath, other RF sources).

Position & Height Localisation

IndoTrack (2017) is the foundational device-free CSI tracking paper in this space; more recent work narrows toward multi-person settings, e.g. Shen et al.'s time-selective RNN for busy-room detection (2023). Reported update rates vary by an order of magnitude across otherwise comparable commodity setups, from 10 Hz (Shen et al., capped by their router) to a reported 100 Hz (IndoTrack, unconfirmed against the primary paper), an additional latency risk on top of the lack of height output.

Speaker Identification

Wi-Fi CSI reflections from mouth/lip movement during speech are detectable using only commodity Wi-Fi: Wang et al.'s WiHear (2014) uses directional antennas to sense multiple people's talk simultaneously via MIMO.

Pros

- Effectively free: reuses existing Wi-Fi infrastructure.
- Multi-person and multi-room literature exists (Time-Selective RNN).
- Device-free output with no recognisable imagery.
- The same commodity hardware also has a literature precedent for detecting mouth-movement reflections for speaker identification (WiHear).

Cons

- No published height-estimation precedent at all, the single biggest gap against RQ1.
- Sensitive to environmental changes (furniture, multipath, other RF sources), the primary limit on stability.
- No demonstrated real-time embedded deployment in the sources reviewed.

2.4.1.4. Ultrasonic Ranging

Active ultrasonic ranging measures distance via time-difference-of-arrival (TDOA) or time-of-flight on emitted pulses, sidestepping the "room full of talking people" interference problem a passive-listening approach would face.

Segers et al. (2015) track up to four transmitters at 1-2cm accuracy on an FPGA using frequency-shift-keyed ultrasound. Segers et al. do not report an update rate or latency figure, so latency here is an open question rather than a documented one.

Pros

- Sidesteps the "room full of talking people" interference problem passive listening would face.
- Demonstrated real-time operation on embedded hardware (an FPGA, per Segers et al.).
- Reasonable literature base for ranging and positioning.

Cons

- No gathered source addresses height or elevation.
- Full coverage needs multiple synchronised nodes, reintroducing multi-sensor complexity.
- Custom-built hardware, roughly €200-400 (rough estimate, not a single purchasable SKU).

2.4.1.5. Infrared / Thermal

Low-resolution thermal/thermopile arrays measure a grid of per-pixel temperature readings; a person would appear as a warm blob against the cooler background, and the same low pixel count that limits resolution and range is itself a privacy advantage, since it's far below what's needed to identify a person or resolve recognisable imagery.

Most directly relevant to RQ1, Naser et al. (2021) explicitly extend thermal-array distance estimation to height measurement using a neural network, and should be read first. Feasibility on embedded hardware is demonstrated by Shubha and Shastrimath (2022), who report 99%

occupancy accuracy on a Grid-EYE/AMG8833 array on a Raspberry Pi, and Klir et al. (2025) extend the same class of Grid-EYE sensor to multi-person localisation (relevant to RQ4). Naser et al.'s sensor supports a 0.5-64 Hz sampling rate; the specific Grid-EYE/AMG8833 breakout identified for procurement is coarser, exposing only a 10 Hz or 1 Hz frame-rate register.

Sensor hardware cost is in the order of 2 x €40-55 excluding the embedded compute platform, confirmed at €49.60 (incl. VAT) via [Kiwi Electronics](#) (NL), and the I2C interface is straightforward to bring up. That price is for the AMG8833's narrow ~60° field of view, sufficient only for a small room; wider coverage without narrowing the FoV would normally mean mounting overhead, which is not an option for the Holobox's fixed installation, so room-scale coverage instead means the wider-FoV MLX90640 (110° FoV) variant, €75.01 (incl. VAT) at [Kiwi Electronics](#) (NL). Wherever thermal is paired with a room-scale primary sensor in [Combined Sensor Approaches](#) below, two units are needed for full room coverage rather than one; since coverage is then split between two sensors, neither one needs the wider (pricier) FoV on its own, so two of the cheaper narrow-FoV AMG8833 units (roughly €100 total) are the cost basis used there, not the single wide-FoV part.

Pros

- Height estimation is demonstrated directly from sensor data, not just a proxy (Naser et al.).
- Low sensor-only cost (€40-55 narrow-FoV; €70-75 for the wide-FoV variant needed for room-scale coverage).
- Demonstrated real-time embedded operation on a Raspberry Pi.
- Low pixel count doubles as a privacy strength: output cannot resolve recognisable imagery.

Cons

- Effective range constrains it to smaller rooms unless the wider-FoV variant is used.
- Lower spatial resolution overall, which needs checking against the Holobox's actual dimensions.

2.4.1.6. Capacitive & Floor-based Sensing

Capacitive / Electric-Field Sensing

Two device-free variants exist: proximity-plate sensing (measuring distance to a body via a change in capacitance) and floor-embedded capacitive electrodes (requiring a person to stand on the sensed area). Bian et al.'s survey (2024) covers the design space broadly, and Faulkner et al.'s CapLoc (2020) is a floor-embedded mat evaluated in the EvAAL indoor-localisation competition. Range is the main constraint: plate sensors top out around 0.3-1m, so whole-room coverage needs a floor grid or many nodes, and the signal is inherently presence/proximity rather than spatial, so it is not naturally suited to a height reading at all.

Floor-based (Pressure) Sensing

Floor-based sensing (pressure, piezoresistive, or capacitive-floor mats) is inherently a 2D floor-

plane measurement, but because each footstep is a spatially discrete event, it naturally disambiguates multiple people. Gatto et al. (2015) use a 64x64 pressure-switch array per tile, and Ibara et al. (2013) show gait-based individual identification from stride length and direction.

Pros

- Floor-based variants trivially solve "how many people, where."
- Solid literature base for both sub-variants.
- Inherently non-identifying (presence/proximity or pressure, not imagery).

Cons

- No height signal at all from either sub-variant.
- Capacitive sensing has short range, needing a dense grid for whole-room coverage.
- Floor-based sensing requires physical installation under the existing floor; cost can run into the thousands of euros.

2.4.1.7. Laser Doppler Vibrometry

Speaker identification only; not evaluated for RQ1 position/height localisation.

A laser Doppler vibrometer (LDV) measures skin-surface vibration at a distance via Doppler shift, picking up throat vibration during speech even in noisy conditions; Avargel and Cohen (2011) combine an LDV sensor aimed at the larynx with a standard microphone to enhance speech quality under transient noise.

2.4.1.8. Microphone Array (Passive Acoustic)

Speaker identification only; not evaluated for RQ1 position/height localisation.

Berghi and Jackson (2023) use a microphone array with a convolutional recurrent neural network to localise an active talker's position from multichannel audio alone, requiring no camera or radar hardware at all.

2.4.2. Combined Sensor Approaches

The individual technologies above are also weighed as sensor combinations, where pairing two of them addresses a gap neither has alone.

2.4.2.1. 2D LiDAR + Thermal Fusion

As with the other thermal-fusion options above, the thermal side here is two AMG8833 units (~€100 total), splitting room coverage between them rather than relying on one pricier wide-FoV MLX90640 or a single narrow-FoV unit that falls short on its own. The representative LiDAR module itself, the RPLIDAR A2M8, also now shows as discontinued/out of stock at major distributors (e.g. Seeed Studio), an added sourcing risk on top of the concerns below.

Pros

- 2D LiDAR contributes strong in-plane position accuracy and occlusion tolerance (Bouazizi et al.).
- Thermal contributes the same direct height-estimation precedent used elsewhere in this chapter (Naser et al.).

Cons

- Sensor-only cost of €350-390, driven up by needing two thermal units.
- Combining two independent position-sensing streams from scratch introduces synchronisation and fusion complexity flagged as a disproportionate implementation risk for this project's timeline, as described in the Project Plan (Chapter 4.3, Timeline).
- The representative RPLIDAR A2M8 module is discontinued at major distributors, an unresolved sourcing risk specific to this combination.
- Does not reuse an already-working single-sensor pipeline; both position-sensing streams are new to this combination.

2.4.3. Comparison Summary

Table 3. Candidate Sensor Comparison Summary

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
mmWave Radar	TI IWR6843AOPEVM (see Module Selection for the variant comparison)	€160-185	Yes, keypoint-based	10-20 Hz
Infrared/Thermal Array	AMG8833 breakout, Kiwi Electronics (NL)	€40-55	Yes, direct	1 Hz or 10 Hz
2D LiDAR + Thermal (sensor fusion)	RPLIDAR A2M8 (discontinued at major distributors) + 2x AMG8833, Kiwi Electronics (NL)	€350-390	Partial, via fusion	5-15 Hz
Wi-Fi CSI Sensing	None. It reuses existing commodity Wi-Fi hardware	Approx. €0 (existing infrastructure)	None found	10-100 Hz
Ultrasonic Ranging	Custom transducer array (e.g. SPM0404UD5 MEMS elements, per Segers et al. (2015))	€200-400 (rough estimate for custom-built hardware, not a single purchasable SKU)	None found	Not reported
Capacitive & Floor-based Sensing	Custom-built capacitive/pressure floor mat. There is no off-the-shelf module	€80-2000+, depending on variant (rough estimate for custom-built hardware)	None found	Not reported

Figures above are sensor/module hardware only. All candidates other than Wi-Fi CSI also need an embedded compute platform to run the sensing pipeline; that cost is excluded here since it is a shared requirement across candidates rather than something that distinguishes one sensor choice from another. Latency figures are the sensor's own documented output rate, not a measured end-to-end pipeline latency; see the caveat already noted in Evaluation Criteria.

Table 4. Criteria Comparison Matrix

Technology	Height Est.	Embedded Real-Time	Multi-Person	Fusion-Ready	Privacy	Literature	Price Range (sensor only)
mmWave Radar	✓	✓	✓	✓	✓	✓	€160-250
Infrared/Thermal Array	✓	✓	✓	✓	✓	✓	€80-110
2D LiDAR + Thermal Fusion	✓	□	✓	✓	✓	✓	€350-390
Wi-Fi CSI Sensing	□	□	✓	□	✓	✓	~€0 (existing infrastructure)
Ultrasonic Ranging	□	✓	□	□	✓	✓	€200-400
Capacitive & Floor-based Sensing	□	□	✓	□	✓	✓	€80-2000+

2.4.4. Recommended Approach

On the criteria set out earlier, mmWave radar comes out ahead: it is the only approach with keypoint-level pose literature, it extends to the RQ4 speaker-identification problem via the same sensor, and it has the strongest literature base of any candidate reviewed; the infrared/thermal array is cheaper and has a direct height-estimation precedent, but mmWave is the stronger technical choice. This is a recommendation of the modality; the specific IWR6843 variant is addressed in [Module Selection](#).

mmWave is by far the most expensive candidate reviewed, and no project budget has yet been agreed with MindLabs, so that cost is a precondition to clear with the company mentor before ordering hardware. Implementation risk is managed instead by building the pipeline one capability at a time (position, then height, then multi-person clustering, then speaker detection), each on an already-working previous phase.

2.5. Module Selection

Candidate Sensor Technologies recommends mmWave radar as the sensing modality, built around the TI IWR6843 chip. The chip ships on several evaluation-board variants at different price points and field-of-view characteristics, given below as azimuth (angular coverage side-to-side) / elevation (angular coverage top-to-bottom, i.e. floor-to-head reach). This chapter compares those variants directly and picks one. The retrieved product listings are current as of 07-09-2026 and may vary when checking on a different date.

- **IWR6843ISK**: the standard long-range antenna plug-in module, €248,23 at current **Mouser NL** listings, with a 120° azimuth / a narrower 30° elevation field of view tuned for range over coverage, which works against a room-scale height-estimation use case rather than for it.
- **IWR6843ISK-ODS**: a wide-field-of-view (120° azimuth / 120° elevation) antenna variant, etched on a Rogers substrate, that TI markets specifically for overhead/ceiling-mounted people counting and tracking (~12 m range), €205,44 at current **Mouser**.
- **IWR6843AOPEVM/AWR6843AOPEVM**: an antenna-on-package (AoP) variant, with the antenna integrated directly into the chip package rather than etched onto a separate PCB, €174,92 at current **Mouser NL/AWR6843AOPEVM**. Matches the ISK-ODS's wide 120° azimuth / 120° elevation field of view in a smaller (~65×60mm).

Table 5. mmWave EVM Variant Comparison

Feature	IWR6843ISK	IWR6843ISK-ODS	IWR6843AOPEVM/AWR6843AOPEVM
Price (€)	248,23	205,44	174,92
Antenna	60-64GHz	60-64GHz	60-64GHz
Rx/Tx	4/3	4/3	4/3
Azimuth Angular Resolution (deg)	15	29	29
Azimuth FOV (deg)	120	120	120
Elevation Angular Resolution (deg)	58	29	29
Elevation FOV (deg)	30	120	120
Max Distance for child/adult (+/-m)	50/60	40/48	41/49

NOTE

The maximum distance for the TI modules is given by the TI **mmWave Sensing Estimator**, configured for the short-range default; it is therefore theoretical, not a measured figure.

2.5.1. Recommendation

Of the three variants in [Table 5](#), the IWR6843AOPEVM (equivalently, AWR6843AOPEVM) is recommended. Elevation, not azimuth or range, is the deciding factor: elevation FOV and resolution determine how much of a person's vertical extent falls inside the beam, the precondition for the keypoint-based height estimate, while every variant already covers the room's short range with margin to spare.

IWR6843ISK is ruled out: its 30° elevation FOV is by far the narrowest of the three, its elevation resolution the coarsest, and it is also the most expensive at €248,23. IWR6843ISK-ODS matches AOPEVM's elevation FOV and resolution exactly, but costs 17% more and its antenna is built for ceiling mounting, an assumption the Holobox has not committed to. AOPEVM reaches the same elevation performance in a smaller, mounting-agnostic form factor at the lowest price of the three, which is why it is carried forward into the phased mmWave implementation in the Conclusion.

2.6. Conclusion

Candidate Sensor Technologies and Module Selection recommend mmWave radar, specifically the IWR6843AOPEVM: the only approach with keypoint-level pose literature for height estimation, non-identifying output, and the strongest literature base of any candidate reviewed, whose antenna-on-package design also gives the elevation coverage needed for room-scale height at the lowest price of the variants compared.

Implementation risk is managed by building the mmWave pipeline in capability-based phases (position, then height, then multi-person clustering, then speaker detection), each on an already-working previous phase; mmWave remains the most expensive candidate reviewed, so mentor approval of its cost is a precondition, not a detail to resolve later. Sensor fusion (mmWave or 2D LiDAR with a thermal sensor) was deliberately not adopted as the starting recommendation: mmWave alone already has a plausible path to a height estimate, none of the fusion combinations has been tested together in the literature, and the thermal side needs two units to match mmWave's coverage, adding real cost rather than a marginal one. Fusion remains a documented fallback if the single-sensor result underperforms, not a precondition for starting.

3. Bibliography

- Alam, M. A. U., Rahman, M. M., & Widberg, J. Q. (2021). *PALMAR: Towards Adaptive Multi-inhabitant Activity Recognition in Point-Cloud Technology*. <https://arxiv.org/pdf/2106.11902>
- Avargel, Y., & Cohen, I. (2011). *Speech measurements using a laser Doppler vibrometer sensor: Application to speech enhancement*. Joint Workshop on Hands-free Speech Communication and Microphone Arrays (HSCMA), Edinburgh, Scotland. <https://israelcohen.com/wp-content/uploads/2018/05/HSCMA2011.pdf>
- Berghi, D., & Jackson, P. J. B. (2023). *Audio Inputs for Active Speaker Detection and Localization via Microphone Array*. <https://arxiv.org/pdf/2307.14739>
- Bian, S., Liu, M., Zhou, B., Lukowicz, P., & Magno, M. (2024). *Body-Area Capacitive or Electric Field Sensing for Human Activity Recognition and Human-Computer Interaction: A Comprehensive Survey*. arXiv. <https://arxiv.org/pdf/2401.06000>
- Bouazizi, M., Mora, A. L., Feghoul, K., & Ohtsuki, T. (2024). *Activity Detection in Indoor Environments Using Multiple 2D Lidars*. Sensors. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11154263/>
- Faulkner, N., Parr, B., Alam, F., Legg, M., & Demidenko, S. (2020). *Device Free Localization with Capacitive Sensing Floor (CapLoc)*. IEEE Sensors Applications Symposium (SAS). <https://www.researchgate.net/publication/344620665>, building on the earlier CapFloor system by Braun, Heggen, & Wichert (2011).
- Gatto, A., Bassoli, E., Contigiani, M., Frontoni, E., Mancini, A., & Zingaretti, P. (2015). *Energy Harvesting Smart Floor for Indoor People Localization and Tracking*. ASME International Design Engineering Technical Conferences (IDETC-CIE). <https://www.researchgate.net/publication/314616906>
- Huang, X., Cheena, H., Thomas, A., & Tsoi, J. K. P. (2021). *Indoor Detection and Tracking of People Using mmWave Radar*. Journal of Sensors, 2021, Article 6657709. <https://onlinelibrary.wiley.com/doi/10.1155/2021/6657709>
- Ibara, K., Hirakawa, M., & Kanetsuna, K. (2013). *Identifying Individuals' Footsteps Walking on a Floor Sensor Device*. Active Media Technology (AMT 2013), Lecture Notes in Computer Science. <https://www.researchgate.net/publication/295102931>
- Klir, S., Lerch, J., Benkner, S., & Khanh, T. Q. (2025). *Multi-Person Localization Based on a Thermopile Array Sensor with Machine Learning and a Generative Data Model*. Sensors, 25(2), 419. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11769272/>
- Li, X., Zhang, D., Lv, Q., Xiong, J., Li, S., Zhang, Y., & Mei, H. (2017). *IndoTrack: Device-Free Indoor Human Tracking with Commodity Wi-Fi*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 1(3), 1-22.
- Miller, R. B. (1968). *Response time in man-computer conversational transactions*. Proceedings of the AFIPS Fall Joint Computer Conference, 33, 267-277.
- Naser, A., Lotfi, A., & Zhong, J. (2021). *Towards human distance estimation using a thermal sensor array*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8204928/>

- Ngamakeur, K., Yongchareon, S., Yu, J., & Ur Rehman, S. (2020). *A Survey on Device-free Indoor Localization and Tracking in the Multi-resident Environment*. ACM Computing Surveys, 53(4). <https://dl.acm.org/doi/fullHtml/10.1145/3396302>
- Nielsen, J. (1993). *Usability Engineering*. Academic Press. Response-time limits (0.1 s, 1.0 s, 10 s) also summarised by the Nielsen Norman Group: <https://www.nngroup.com/articles/response-times-3-important-limits/>
- Pegoraro, J., & Rossi, M. (2022). *Human Tracking with mmWave Radars: a Deep Learning Approach with Uncertainty Estimation*. <https://arxiv.org/pdf/2205.03274>
- Segers, L., Van Bavegem, D., De Winne, S., Braeken, A., Touhafi, A., & Steenhaut, K. (2015). *An Ultrasonic Multiple-Access Ranging Core Based on Frequency Shift Keying Towards Indoor Localization*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4570340/>
- Shen, L.-H., Hsiao, A.-H., Chu, F.-Y., & Feng, K.-T. (2023). *Time-Selective RNN for Device-Free Multi-Room Human Presence Detection Using WiFi CSI*. <https://arxiv.org/pdf/2304.13107>
- Shubha, B., & Shastrimath, V. V. D. (2022). *Real-Time Occupancy Detection System Using Low-Resolution Thermopile Array Sensor for Indoor Environment*. IEEE Access, 10, 130981-130995. <https://www.researchgate.net/publication/366350707>
- Su, Y., Hou, H. C., Lan, H., & Ma, C. Z.-H. (2024). *A High-Fidelity mmWave Radar Dataset for Privacy-Sensitive Human Pose Estimation* (the mmFree-Pose dataset). Bioengineering. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12383697/>
- Wang, G., Zou, Y., Zhou, Z., Wu, K., & Ni, L. M. (2014). *We Can Hear You with Wi-Fi!* (WiHear). Proceedings of the 20th Annual International Conference on Mobile Computing and Networking (MobiCom).
- Xu, C., Li, Z., Zhang, H., Rathore, A. S., Li, H., Song, C., Wang, K., & Xu, W. (2019). *WaveEar: Exploring a mmWave-based Noise-resistant Speech Sensing for Voice-User Interface*. Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys), 14-26.
- Zhu, B., He, Z., Xiong, W., Ding, G., Huang, T., & Xiang, W. (2024). *ProbRadarM3F: mmWave Radar based Human Skeletal Pose Estimation with Probability Map Guided Multi-Format Feature Fusion*. <https://arxiv.org/html/2405.05164v2>